

3.Subqueries

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0.0.1 Definitions

What Are Subqueries?

Queries embedded into other queries

Relational databases store data in multiple tables

Subqueries merge data from multiple sources together

Helps with adding other filtering criteria

Example

Need to know the region each customer is from who has had an order with freight over 100

```
SELECT DISTINCT customerID
FROM Orders
WHERE Freight > 100
```

	customerID
1	RICSU
2	ERNSH
3	FRANK
4	WARTH
5	MORGK
6	QUICK
7	RATTC
8	HUNGO
9	GODOS
10	OLDWO
11	SAVEA
12	SPLIR
13	BONAP
14	LEHMS
15	PICCO
16	SEVES
17	BLONP
18	QUEEN
19	BERGS
20	HILAA
21	MAGAA
22	MEREP
23	SUPRD
24	LILAS
25	VAFEE

```
SELECT
CustomerID
,CompanyName
,Region
FROM Customers
WHERE customerID IN ('RICSU'
, 'ERNSH'
, 'FRANK'
, 'WARTH'
, 'MORGK'
, 'QUICK'
, 'RATTC'
, 'HUNGO'
, 'GODOS'
, 'OLDWO'
. . .)
```

Working with Subquery Statements

Always perform the innermost
SELECT portion first

```
SELECT
CustomerID
,CompanyName
,Region
FROM Customers
WHERE customerID IN (SELECT
customerID
FROM Orders
WHERE Freight > 100 );
```

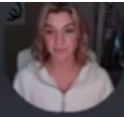
DBMS is performing two operations

1. Getting the order numbers for the product selected
2. Adding that to the WHERE clause and processing the overall SELECT statement

Putting At All Together

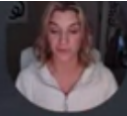
```
SELECT Customer_name,
Customer_contact
FROM Customers
WHERE cust_id IN
(SELECT customer_id
FROM Orders
WHERE order_number IN
(SELECT order_number
FROM OrderItems
WHERE prod_name =
'Toothbrush');
```

Name and contact of customers
with a toothbrush order



Customer name	Customer contact
John Smith	johnsmith@example.com
Alice Johnson	alice@example.com
Mary Clark	mary@example.com
John Smith	johnsmith@example.com

Subqueries for Calculations



Total number of orders placed by every customer

Customer_name	Customer_state	Orders
Becky	IA	5
Nita	CA	6
Raj	OH	0
Steve	AZ	1

```
SELECT customer_name
, customer_state
, (SELECT COUNT (*) AS orders
   FROM Orders
   WHERE orders.customer_id =
customer.customer_id) AS orders
FROM customers
ORDER BY customer_name
```

Best Practices with Subqueries

There is **no limit** to the number of subqueries you can have

Performance slows when you nest too deeply

Subquery selects can only **retrieve a single column**

0.0.2 Joins